

7448 AI

Overview

7448 AI is a biotechnology-driven artificial intelligence (AI) project pioneering the intersection between gut microbiome science and neural network-based intelligence. Our mission is to revolutionize how clinicians, researchers, and physicians understand the gastrointestinal (GI) system: one of the most complex, yet underexplored physiological systems.

We are developing an AI system that solves a fundamental bottleneck in GI diagnostics. When physicians order microbiome testing today, they learn which bacteria are present in a patient's gut; they do not (specifically) know what those bacteria are doing and what they are responsible for in the patient's GI system. By using sequential DNA input, our AI system bridges this disparity by predicting the complete functional output of the gut microbiome: the thousands of metabolites that microbial communities produce.

Our technology will transform gut microbiome analysis from descriptive taxonomy into functional medicine. For example, instead of telling a clinician that certain bacterial species are depleted, we report that the microbiome is producing insufficient levels of anti-inflammatory compounds while generating elevated levels of pro-inflammatory metabolites. This perspective explains why patients experience symptoms and it allows us to develop personalized recommendations for interventions (dietary, probiotic, and more) to correct these metabolic imbalances.

The distinction is profound. Knowing that specific bacteria are missing is interesting, but it leaves clinicians uncertain on their next steps in treatment. Knowing that protective metabolites are low and harmful metabolites are high immediately points towards therapeutic action. We transform ambiguous taxonomic data into clear metabolic assessments with evidence-based intervention strategies.

The scientific challenge we're addressing is learning an extraordinarily complex biological translation function. The gut microbiome operates as an integrated biochemical network where hundreds of bacterial species interact, compete, and cooperate to transform dietary inputs into metabolic outputs. These metabolites (e.g. butyrate, propionate, trimethylamine, indole derivatives, and modified bile acids) serve as the interface between the microbiome and human health. In particular, they modulate inflammation, regulate metabolism, influence neurotransmitter production, and communicate with the immune system. Currently, measuring these metabolites requires expensive, time-consuming mass spectrometry that isn't practical for routine clinical use. We're building machine learning (ML) models that learn to predict metabolite production from readily available sequencing data.

With this initiative's success, we can expect to see personalized gut health interventions, accelerate therapeutic development by allowing computational screening of treatments before clinical trials, and provide early warning systems for detecting metabolic imbalances before they manifest as disease.

The Scientific Problem

Today's microbiome testing relies primarily on 16S ribosomal RNA sequencing or metagenomic shotgun sequencing. These methods excel at answering one question: which bacteria are present? They generate detailed taxonomic profiles showing the relative abundance of bacterial species in a patient's gut. While

scientifically valuable, this presents a critical limitation for clinical practice: taxonomy does not predict function. Two patients may harbour the same bacterial species yet experience entirely different health outcomes. Bacterial function depends on strain-level, genetic variations, microbial community interactions, dietary context, and environmental conditions. A bacterium that produces beneficial compounds in one context might generate inflammatory metabolites in another.

Current diagnostic approaches measure the wrong thing. What matters for human health is not primarily which microbes are present, but what those microbes are doing specifically, what metabolites they produce. These metabolites mediate nearly all microbiome-host interactions:

- Short-chain fatty acids (butyrate, propionate, acetate) nourish intestinal cells and regulate inflammation
- Bile acid metabolites influence metabolism and immune function
- Trimethylamine and derivatives affect cardiovascular health
- Tryptophan metabolites impact mood and neurological function

The gold standard for measuring these metabolites is mass spectrometry which is a technique that is expensive (\$500-2000 per sample), time-intensive (days to weeks), technically demanding, and therefore impractical for routine clinical use.

Complexity: Understanding the relationship between microbial composition and metabolic output is extraordinarily complex.

Non-linear interactions: Metabolite production emerges from intricate networks of microbial cooperation and competition. A single metabolite may be produced by multiple species or modified through multi-step pathways involving different bacteria.

High dimensionality: The gut contains hundreds to thousands of bacterial species producing hundreds to thousands of distinct compounds. Mapping this relationship requires learning a transformation with millions of parameters.

Individual variability: The composition-to-function relationship varies between individuals based on diet, medications, host genetics, and disease state.

Data challenges: Both sequencing and metabolomics generate noisy data with batch effects and missing values.

Clinical consequence: This functional gap creates a substantial barrier to clinical translation. When a physician sees depleted beneficial bacteria or elevated pathogenic species, they face uncertainty: What does this mean for my patient's symptoms? What intervention should I recommend? Without functional insight, microbiome testing remains descriptive rather than actionable.

Our Solution: We address these concerns by training deep learning models to learn the complex mapping from microbial composition (readily measured via DNA sequencing) to functional metabolic output (traditionally requiring mass spectrometry). By learning from paired microbiome-metabolome datasets, our neural networks capture the biochemical relationships that govern metabolite production, transforming microbiome analysis from a descriptive exercise into an actionable diagnostic tool.

Machine Learning Architecture: Learning Biological Function

With our training data assembled, the core scientific challenge becomes designing machine learning architectures that can effectively learn the complex relationship between microbiome composition and metabolite production. This requires careful thought about how we represent the data, what neural network architectures we employ, and how we train and regularize the models to learn generalizable patterns rather than memorizing noise. Let's start by thinking about data representation, because how we encode our inputs and outputs substantially affects what the model can learn.

For the metagenomic input, we have several representation choices, each with different tradeoffs. The simplest representation is a taxonomic abundance vector where each dimension represents a bacterial species or genus, and the value indicates its relative abundance in the sample. This might be a vector with several thousand dimensions corresponding to all possible microbial taxa. This representation directly captures community structure but doesn't explicitly encode functional information.

A more sophisticated representation uses functional gene abundance, where we quantify not which species are present but which metabolic genes and pathways are encoded in the community. After sequencing, we can use computational tools to map sequences to gene families and metabolic pathways regardless of which species carry them. This gives us a vector indicating the abundance of genes for various metabolic functions like carbohydrate fermentation, amino acid synthesis, bile acid transformation, and vitamin production. This representation more directly captures metabolic potential, which is often more predictive of actual metabolite production than taxonomy alone.

The most informative representation combines both taxonomic and functional information in a single large feature vector. This allows the model to learn both direct functional relationships and also capture cases where the specific bacterial species matters beyond just the genes it carries. For instance, the same metabolic genes might be expressed differently depending on which species harbors them, or certain species might have regulatory mechanisms that affect when genes are active.

We might also explore more sophisticated representations borrowed from natural language processing. We could treat metagenomic sequences as if they were sentences, where each gene or genomic region is like a word, and use transformer architectures to learn contextual embeddings that capture relationships between different genetic elements. This is more computationally intensive but could potentially capture subtle sequence-level patterns that simpler representations miss.

For the output representation, we encode metabolomic data as a vector where each dimension corresponds to a specific metabolite and the value represents its concentration or abundance. The dimensionality might range from hundreds to thousands depending on how comprehensive our metabolomics measurements are. One important consideration is that metabolite concentrations span many orders of magnitude, so we typically work with log-transformed values to put them on a more comparable scale.

The neural network architecture itself requires careful design. A reasonable starting architecture would be a deep feedforward network, sometimes called a multilayer perceptron. This consists of an input layer that accepts the metagenomic feature vector, several hidden layers that progressively transform the representation, and an output layer that produces metabolite predictions. Each hidden layer applies a linear

transformation followed by a non-linear activation function, allowing the network to learn complex non-linear relationships.

The number and size of hidden layers represents a key architectural choice. Deeper networks with more layers can learn more complex functions but are harder to train and more prone to overfitting. A typical architecture might have three to five hidden layers, with the number of neurons in each layer gradually decreasing as we move from input to output. For instance, if we have five thousand input features and one thousand output metabolites, we might use hidden layers with three thousand, two thousand, one thousand, and five hundred neurons respectively. This creates a bottleneck that forces the network to learn compressed representations.

Between hidden layers, we would include several components that help with training stability and generalization. Batch normalization layers normalize the inputs to each layer, which stabilizes training and allows us to use higher learning rates. Dropout layers randomly deactivate some neurons during training, forcing the network to learn redundant representations and preventing it from relying too heavily on any particular pathway. This acts as a powerful regularizer against overfitting.

More sophisticated architectures incorporate attention mechanisms that learn which input features are most relevant for predicting each metabolite. We could implement multi-head attention layers that learn to focus on different subsets of microbial features when predicting different metabolites. This not only might improve prediction accuracy but also provides interpretability by showing us which microbial features the model considers important for each metabolic prediction.

Another powerful architectural pattern is the encoder-decoder structure. The encoder portion takes the metagenomic input and compresses it into a lower-dimensional latent representation that captures the essential metabolic state of the microbiome. The decoder then expands this latent representation to predict the full metabolomic profile. The latent representation becomes valuable in its own right as a learned feature we can analyze and use for other purposes like clustering samples with similar metabolic states.

We should also consider multi-task learning architectures where the model learns to predict not just metabolites but also other related outcomes simultaneously. We might train the model to predict metabolites, clinical outcomes like disease state, and perhaps even intermediate quantities like predicted enzyme activities. By learning multiple related tasks together, the model often develops richer internal representations that generalize better. The intuition is that being forced to use the same learned features for multiple purposes encourages the model to learn more fundamental biological patterns.

The loss function we optimize during training critically shapes what the model learns. For continuous metabolite predictions, mean squared error is a natural starting point, penalizing the model proportionally to the squared difference between predictions and true values. However, given that metabolite concentrations span many orders of magnitude, we typically compute loss on log-transformed values to give equal relative importance to all metabolites regardless of their absolute abundance. We might also use weighted losses that place higher importance on metabolites that are particularly clinically relevant. If certain metabolites like butyrate or secondary bile acids are known to be especially important for health outcomes, we can weigh their prediction errors more heavily in the loss function. This biases the model toward learning to predict these important metabolites more accurately, potentially at the expense of less clinically relevant compounds.

Regularization strategies beyond dropout are important for preventing overfitting given the high dimensionality of our problem. L2 regularization adds a penalty term to the loss that discourages large parameter values, encouraging the model to learn simpler functions. We might also employ early stopping, where we monitor performance on a validation set during training and stop when validation performance starts degrading even if training performance continues improving.

The training process itself involves feeding the model batches of training examples, computing predictions and loss, calculating gradients using backpropagation, and updating parameters using an optimizer like Adam that adapts learning rates for each parameter. We would train for many epochs, each epoch consisting of one complete pass through the training data. Throughout training, we monitor both training and validation loss to ensure the model is learning and not overfitting.

Learning rate scheduling helps the model converge to better solutions. We might start with a relatively high learning rate to make rapid progress, then gradually decrease it as training progresses so the model can fine-tune its parameters. We could also use techniques like learning rate warm-up where we start with a very small learning rate and gradually increase it during the first few epochs.

Data splitting requires care to avoid information leakage. We divide our data into training, validation, and test sets, typically using something like a seventy-fifty-fifty split. Critically, if we have multiple samples from the same person, all samples from that person must go into the same split. Otherwise the model might effectively memorize person-specific patterns rather than learning generalizable biological relationships.

We would implement cross-validation during hyperparameter tuning to get robust estimates of model performance. In k-fold cross-validation, we divide the training data into k groups and train k different models, each using a different group as its validation set. This gives us k independent estimates of performance that we can average.

For hyperparameter optimization, we need to systematically explore choices like network architecture, learning rate, dropout rate, batch size, and regularization strength. We might use grid search to try all combinations, random search to sample the hyperparameter space more efficiently, or Bayesian optimization to intelligently explore based on previous results. Each configuration is evaluated using cross-validation, and we select the configuration that performs best on average.

Validation Framework: Ensuring Scientific Rigor

Building a model that performs well on training data is relatively straightforward. The challenge is ensuring it generalizes to new data and has actually learned biologically meaningful relationships rather than memorizing spurious correlations. This requires a comprehensive validation framework that examines model performance from multiple angles.

The foundation of validation is evaluating prediction accuracy on held-out test data that the model has never seen during training or hyperparameter tuning. For each metabolite, we compute metrics like Pearson correlation between predicted and true concentrations, mean absolute error, and mean squared error. These metrics quantify how well the model captures both the relative ordering of samples and the absolute magnitude of metabolite levels.

However, aggregate metrics across all metabolites and samples hide important nuances. We should break down performance by metabolite class to understand which types of compounds the model predicts well versus poorly. Are short-chain fatty acids predicted more accurately than secondary bile acids? Are amino acid derivatives easier than complex lipids? These patterns help us understand what aspects of microbial metabolism the model has captured.

We should also examine whether prediction accuracy varies across sample types and patient populations. Does the model work as well for samples from patients with inflammatory bowel disease as for healthy controls? Does performance hold across different age groups, dietary patterns, or geographic populations? Finding that accuracy drops for certain subgroups indicates our training data may not adequately represent those populations.

An important validation is testing the model's ability to predict metabolic changes in response to interventions. If we have longitudinal data where the microbiome was perturbed by dietary changes, antibiotics, or probiotics, we can ask whether the model correctly predicts how metabolites change. Given a baseline sample and knowledge of the intervention, can we predict the follow-up metabolite profile? This tests whether the model has learned causal relationships that allow it to simulate interventions.

Biological plausibility checks are essential for ensuring the model hasn't learned spurious patterns. We can examine feature importance to see which metagenomic features most strongly influence each metabolite prediction. Do these make biological sense? If we're predicting butyrate, the model should heavily weight bacteria known to produce butyrate like *Faecalibacterium prausnitzii* or *Roseburia* species. If it's giving high importance to implausible features, this suggests overfitting to training data peculiarities.

Feature importance can be assessed through several methods. We can compute gradient-based importance scores that measure how sensitive each output is to each input feature. We can use permutation importance by randomly shuffling each input feature and measuring how much prediction accuracy drops. We can employ SHAP values that provide theoretically grounded attributions showing how much each feature contributed to each specific prediction.

Ablation studies help us understand what information the model relies on. We train variant models with certain types of features removed and measure how performance degrades. For instance, training a model using only taxonomic features without functional gene information tells us how much functional data adds. Training on single-lab data versus multi-lab data quantifies the benefit of diversity.

We should validate generalization to completely new data sources. The most stringent test is collecting samples from a new laboratory using potentially different protocols, processing them, and evaluating our pre-trained model without any retraining or fine-tuning. If performance holds up, we have strong evidence of genuine generalization. If performance drops substantially, we need to either improve protocol standardization or make our models more robust to protocol variations.

Cross-population validation is similarly important. We might train the model on samples from one geographic region and test on another, or train on healthy individuals and test on disease patients. These experiments reveal whether the model has learned universal biological principles or population-specific patterns.

We should also examine prediction uncertainty and calibration. When the model makes a prediction, how confident should we be that it's accurate? Good uncertainty estimates are valuable clinically because they

tell us which predictions to trust versus which require confirmatory testing. We can assess calibration by checking whether the model's confidence scores actually match empirical accuracy rates.

Error analysis provides crucial insights for model improvement. We should systematically examine cases where predictions are most inaccurate. Are there patterns in which samples or metabolites are hardest to predict? Often this reveals data quality issues like contamination in certain samples, or identifies biological phenomena the model hasn't captured like rare metabolic subtypes.

We would also validate that the model hasn't simply memorized the training distribution but can actually interpolate and extrapolate reasonably. We can create synthetic test cases with microbiome compositions that don't exactly match any training sample and ask whether predictions seem reasonable based on biological knowledge.

Throughout validation, we should be alert for potential pitfalls like batch effects where the model learns to recognize which lab a sample came from rather than learning biology, label leakage where information about outputs accidentally gets encoded in inputs, and selection bias where our training data systematically differs from the target population we want to make predictions for.

Closing Statement

Our work is rooted in a simple belief: the future of gut health is data-driven, intelligent, and deeply personal. Traditional analysis methods often overlook subtle microbial interactions or rare signatures that could signal early-stage conditions. Our model, trained to “read” microbiome data much like a scientist interprets a genome, can detect these intricate relationships and propose patterns unseen by the human eye.

- The 7448 AI Team

The Team

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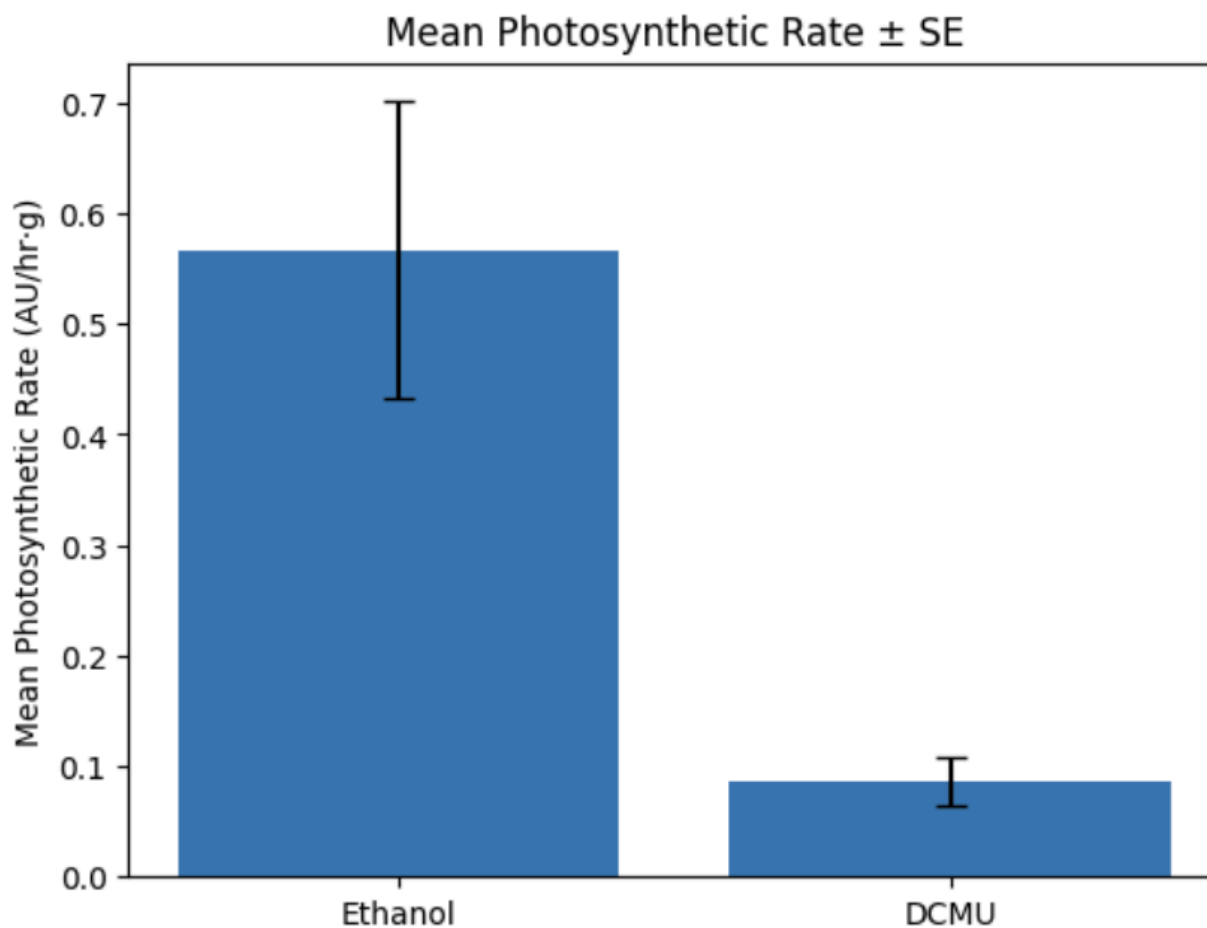


Figure 1. Mean ($\pm$ SE) photosynthetic rates of *Limnophila* sp. measured using bromothymol blue as a CO₂ indicator at 615 nm. Plants were exposed to either an ethanol control or the photosystem II inhibitor DCMU, and photosynthetic rate was calculated from changes in absorbance over time. DCMU-treated samples showed substantially lower photosynthetic rates compared to the ethanol control, indicating inhibition of electron transport and carbon fixation.